GT to WD

By Dr. Jason Lo and Sasank Ravella

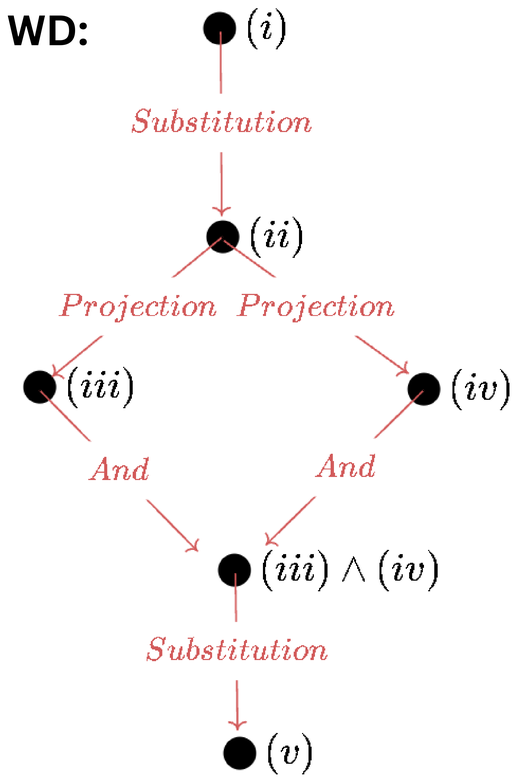
Theorem 1 (Unique Identity): *Let G be a group. Then the identity of G is unique.*

Proof: Suppose that e and f are both identities in G. Then as f is an identity, $ef = e$. But as e is an identity, we also have $ef = f$. Therefore, $e = f$.

Claim	Content
(i)	There are two elements e, f that are identities in G.
(ii)	s.t. $\forall c \in G, \quad ce = ec = c \quad \& \quad cf = fc = c.$
(iii)	$ef = f$
(iv)	$ef = e$
(v)	$e = f$

Diagram	Operation	Formal definition	Intuition
A ↓ B	projection	Suppose claim A is of the form $e_1 \wedge e_2 \wedge \cdots \wedge e_n,$ where each e_i is a claim. Then claim B is a projection of claim A if it is of the form $e_{i_1} \wedge \cdots \wedge e_{i_m},$ where $\{i_1, \ldots, i_m\} \subseteq \{1, \ldots, n\}$.	B is a projection of A if B can be obtained from A by "forgetting" some of the sub-claims in A.
$A_1 \quad \cdots \quad A_n$ ↘ ↙ B	AND	Claim B is the result of the AND operation applied to claims $A_1, \ldots, A_n$ if $BisA_1 \wedge A_2 \wedge \cdots \wedge A_n.$	
A ↓ B	substituition	Claim B is a consequence of claim A by applying a proposition.	

This table shows few operations that we have found in mathematical reasoning.



This is a Wiring diagram that shows the mathematical structure of the proof.

Motivation:

- This project develops a visual method for representing mathematical proofs.
- Shing-Tung Yau's Yau Laboratory Alliance for Mathematics and Artificial Intelligence aims to use AI to generate mathematical conjectures, formalise proofs, and build systems whose reasoning is easier to interpret(1). On the other hand, Navier–Stokes Lost in Translation highlights a crucial limitation with AI: Lean verifies the formal statement and proof submitted to it without checking whether they faithfully represent the original natural-language theorem or argument(2). An AI system may silently alter an assumption, establish a weaker statement, or substitute a different valid proof for the author's reasoning. For the alliance's work, this makes semantic fidelity essential: translations between natural-language claims, Lean proofs, and human-readable explanations must preserve the intended mathematical meaning.

References:

1) https://www.scmp.com/news/china/science/article/3370492/leading-chinese-mathematicians-unite-behind-use-ai-research
2) https://arxiv.org/pdf/2610.08144

CSUN COLLEGE OF SCIENCE AND MATHEMATICS

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Department of Mathematics - California State University, Northridge • Live Oak Hall 1300 • 818.677.2721